School of Biological Sciences | BI3RP3

University of Reading

Impact of Vitamin B12 Supplementation on Neural Activity in the Barrel Cortex of Rats

Uche Dumzo-Ajufo

Supervisor: Ying Zheng

Introduction

Vitamin B12 is essential for neural health, playing a key role in myelination and neurotransmitter synthesis. A deficiency can contribute to cognitive decline and neurological disorders (dos Santos et al., 2024). While its overall effects on the body are well-studied, its direct impact on neural activity remains unclear. The barrel cortex, responsible for processing sensory input from whiskers, serves as a useful model for studying these effects (Petersen, 2007). This study explores how B12 supplementation influences local field potentials (LFPs) in the barrel cortex of neurotypical rats, offering insights into its role in neural function.

Hypothesis & Research Objective

- **Hypothesis:** Vitamin B12 supplementation enhances neural activity in the barrel cortex of rats during whisker pad stimulation.
- **Objective:** To investigate how vitamin B12 supplementation affects the amplitude and latency of local field potentials in the barrel cortex of rats during whisker pad stimulation.

Methodology

Subjects

- **14 adult rats** divided into two groups
  - **B12** Group (n=7): Fed with B12-enriched water.
  - **Control** Group (n=7): Fed with fresh water.
- Rats were anaesthetised.
- Cortical responses to whisker stimulation recorded in vivo using an extracellular microelectrode, local field potentials (LFPs) extracted, data pre-processed.

Experimental Design

- Each rat was tested under **four stimulation intensities**: **0.4, 0.6, 0.8 and 1.2mA**
- Each trial was 2 seconds long and had 10 stimulus pulses (**5 Hz frequency**).
- Key metrics analysed from the **1st** pulse**, 2nd** pulse and **steady-state** (average response from 8th, 9th and 10th pulse). Highlighted in the figure below:

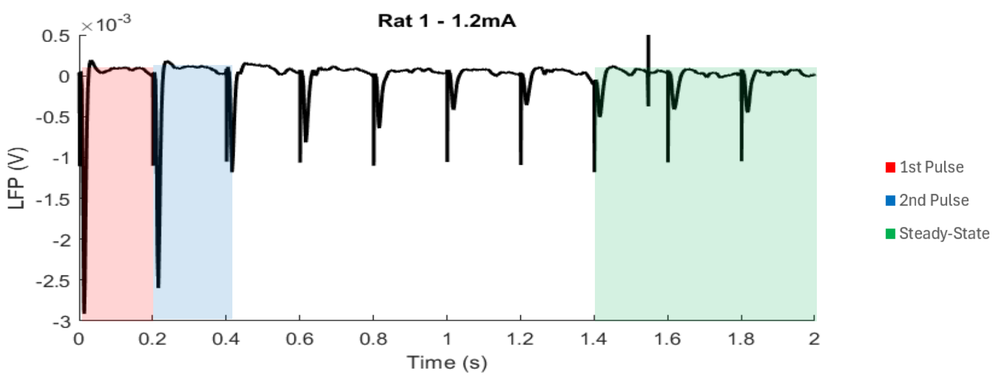


Figure 1. *Evoked LFP response for Rat 1 during the experiment at 1.2 mA in the B12 group, shown as an example. The x-axis represents time in seconds (0–2 s), and the y-axis shows the averaged LFP response amplitude (V) across 50 trials. The data were averaged across all trials to minimise variability and highlight the overall response trend. The first pulse is highlighted in red, the second pulse in blue, and the steady-state response is shown in green.*

Metrics Analysed

- **Peak amplitude**: The highest point of a signal from the baseline.
- **Peak-to peak (P2P) amplitude**: The difference between the highest and lowest points of a signal.
- **Latency:** Time delay in response.

Statistical Analysis

- **Two-way repeated measures ANOVA**: Assessed the effects of group (B12 vs control) and stimulation intensity (0.4-1.2 mA).
- **Post-hoc tests**: Two-sample t-tests were performed to determine which intensities showed significant differences.
- **Effect size (Cohen's d):** Calculated for all significant results to assess the strength of group differences.

References

1. Alruwaili, M., Basri, R., AlRuwaili, R., Albarrak, A.M. and Ali, N.H. (2023). Neurological Implications of Vitamin B12 Deficiency in Diet: A Systematic Review and Meta-Analysis. Healthcare, [online] 11(7), p.958.
2. dos Santos, M.E.R.A., da Silva, K.G., da Silva Souza, A.P., da Silva, A.B.J., da Silva, R.F., da Silva, E.H.A., de Souza, S.L. and Barros, W.M.A. (2024). Relationship between vitamin B12 levels and motor development: A systematic review. Clinical Nutrition ESPEN, [online] 63, pp.148–156..
3. McDougall, S., Vargas Riad, W., Silva-Gotay, A., Tavares, E.R., Harpalani, D., Li, G.-L. and Richardson, H.N. (2018). Myelination of Axons Corresponds with Faster Transmission Speed in the Prefrontal Cortex of Developing Male Rats. eneuro, 5(4), pp.ENEURO.0203-18.2018.
4. Petersen, C.C.H. (2007). The functional organization of the barrel cortex. Neuron, [online] 56(2), pp.339–55.

Results

Average Responses

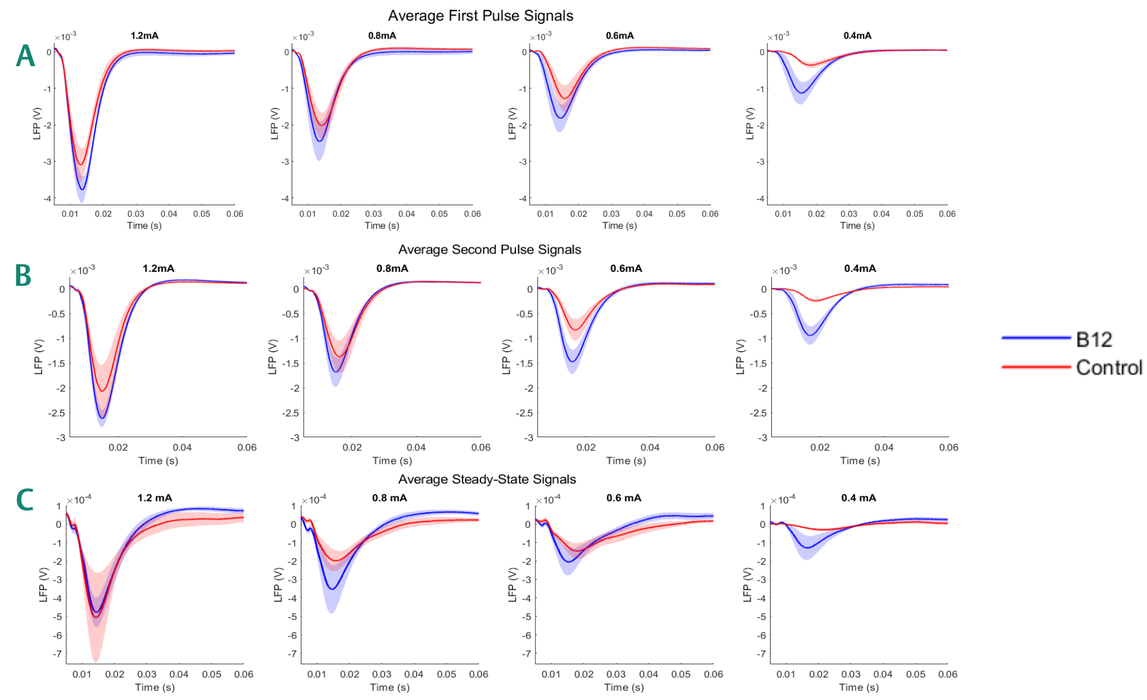


Figure 2. *Average LFP responses across all rats for B12 (blue) and control (red) groups at different stimulation intensities (1.2, 0.8, 0.6, and 0.4 mA). Plots represent the mean LFP response averaged across all rats (n=7 per group) for the first pulse (A), second pulse (B), and steady-state pulse (C). The x-axis shows time (s), and the y-axis indicates LFP amplitude (V). As intensity decreased, peak amplitude decreased. Shaded error bands represent variability (±SEM), highlighting response trends and differences between B12 and control groups at each intensity.*

Two-Way Repeated Measures ANOVA (Main Group Effect)

	Peak	P2P	Latency
1st Pulse	0.057	0.048*	0.127
2nd Pulse	0.013*	0.010**	0.192
Steady-State	0.556	0.651	0.032*

Table 1. *Two-way repeated measures ANOVA results showing main group effects (B12 vs. control) on LFP metrics, rounded to two decimal places. Significant effects are marked with an asterisk (*$p < 0.05$) or double asterisks ($p < 0.01$), indicating statistically significant differences between groups. No significance was found for the interaction effect between group and intensity.*

Post-Hoc Two-Sample t-test

Variable	Intensity	p-value	Cohen's d
1st Pulse Peak	1.2	0.369	
	0.8	0.714	
	0.6	0.352	
	0.4	0.028*	1.33
2nd Pulse Peak	1.2	0.385	
	0.8	0.653	
	0.6	0.077	
	0.4	0.004**	1.92
2nd Pulse P2P	1.2	0.350	
	0.8	0.658	
	0.6	0.080	
	0.4	0.003**	1.94
Steady-State Latency	1.2	0.312	
	0.8	0.633	
	0.6	0.127	
	0.4	0.061	1.11

Table 2. *Post-hoc two-sample t-test results for significant variables at each intensity. The table shows the p-values for the comparisons between the B12 and control groups at the four stimulation intensities for each significant variable. Significant differences were only observed at 0.4 mA. Significant p-values are marked with an asterisk (*), with two asterisks indicating higher significance levels (**$p < 0.01$, *$p < 0.05$). Steady-state latency showed near-significance. Effect sizes (Cohen's d) are also reported for each significant comparison, with larger values indicating a stronger group effect.*

Discussion & Conclusion

- **Summary of Findings:** B12 supplementation increased neural responses at **0.4 mA**, enhancing **1st and 2nd pulse peak amplitude** and **2nd pulse P2P amplitude,** while **reducing steady-state latency**. No effects were seen at 0.6–1.2 mA. **Large effect sizes (d > 1)** indicate a strong effect.
- **Interpretation:** These findings support the hypothesis that B12 enhances neural activity by **amplifying** neural responses. Shorter steady-state latency suggests **faster response times**, possibly due to **improved myelination** or **neurotransmitter function** (McDougall et al., 2018). The lack of effects at higher intensities indicates that B12's impact is **most pronounced under lower stimulation**.
- **Comparison to Previous Studies:** Unlike previous systemic studies, this research examines B12's direct cortical effects, offering new insights into its role in neural modulation.
- **Implications:** B12 may have therapeutic potential for neurodegenerative diseases and sensory processing deficits, highlighting the need for further research into how nutritional interventions might support brain health (Alruwaili et al., 2023).